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Capstone Project Report
on
PLC-Based Sequential Control of Three Mixing Devices Using Timers

Submitted in partial fulfillment of the requirements of the Third Year of
Bachelor of Technology In
Electronics and Computer Science

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Certificate of Approval

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Abstract

Programmable Logic Controllers (PLCs) are widely used in industrial automation to control machines and perform operations according to predefined instructions. This project focuses on the design of a PLC-based sequential control system for three mixing devices, namely Mixer A, Mixer B, and Mixer C, using timers and ladder logic.

The system begins its operation when the Process Start input (X0) is activated. Mixer A, connected to output Y0, starts after a delay of 7 seconds. After Mixer A starts, a further delay of 3.6 seconds is introduced before Mixer B, connected to output Y1, starts. Similarly, Mixer C, connected to output Y2, starts after an additional delay of 5 seconds following the activation of Mixer B. Therefore, Mixer A starts at 7 seconds, Mixer B at 10.6 seconds, and Mixer C at 15.6 seconds after the process begins.

The control system uses PLC timers, input signals, internal relays, and output coils to execute the required sequence. The Process Stop input (X1) or turning OFF the master enable switch terminates the process and switches OFF all three mixers. The control logic is designed to reset the sequence so that the process can be restarted when required.

This project demonstrates the application of PLC-based sequential control in industrial mixing operations. The proposed system reduces manual intervention, improves timing consistency, and provides a structured method for controlling multiple devices. It also provides practical knowledge of ladder logic programming, timer configuration, input/output assignments, and industrial automation.

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CHAPTER 1: OVERVIEW

1.1 Introduction

Industrial automation plays an important role in modern manufacturing and processing industries by improving operational efficiency, accuracy, reliability, and productivity. Programmable Logic Controllers (PLCs) are widely used in industrial automation to control machines, monitor processes, and execute operations according to predefined instructions. PLCs can perform sequential operations using timers, internal relays, input signals, and output devices.

In industrial processing systems, multiple machines often need to operate in a specific sequence with predetermined time intervals. Performing these operations manually requires continuous monitoring and accurate timing by operators. A PLC-based control system can automate these operations, reduce manual intervention, and improve the consistency of the process.

This project focuses on the design of a PLC-Based Sequential Control System for Three Mixing Devices Using Timers. The system consists of three mixing devices, namely Mixer A, Mixer B, and Mixer C. The operation is controlled using a Process Start input (X0), a Process Stop input (X1), and three output signals assigned to the respective mixers.

When the process begins, Mixer A, connected to output Y0, starts after a delay of 7 seconds. After Mixer A starts, Mixer B, connected to output Y1, starts after an additional delay of 3.6 seconds. Similarly, Mixer C, connected to output Y2, starts after a further delay of 5 seconds following the activation of Mixer B. Thus, Mixer A starts after 7 seconds, Mixer B after 10.6 seconds, and Mixer C after 15.6 seconds from the beginning of the process.

The system uses PLC timers and ladder logic to implement the required sequence. Once activated, each mixer remains ON until the Process Stop input is activated or the master enable switch is turned OFF. This ensures that all three mixers stop when the process is terminated.

The project demonstrates the practical application of PLCs in sequential machine control and industrial automation. The proposed system can serve as a basic model for industrial processes that require the coordinated operation of multiple machines at predefined time intervals.

1.2 Problem Statement

In industrial processing systems, multiple mixing devices may need to start at specific time intervals to ensure proper coordination of operations. When these operations are performed manually, operators must monitor the starting time of each machine and ensure that the required delays are maintained. This may increase the workload and lead to timing errors or inconsistent operation.

The problem addressed in this project is to develop a PLC-based control system capable of automatically starting three mixing devices in a predefined sequence using timers.

The system must satisfy the following requirements:

1. Mixer A must start 7 seconds after the process begins.
2. Mixer B must start 3.6 seconds after Mixer A starts.
3. Mixer C must start 5 seconds after Mixer B starts.
4. All three mixers must remain ON after their respective starting times.
5. All mixers must turn OFF when the Process Stop input is activated or the master enable switch is turned OFF.
6. The control sequence must reset so that the process can restart when a new start command is provided.

The proposed solution uses PLC timer instructions, input/output signals, and ladder logic to achieve the required sequential operation. The system is designed to provide a simple and reliable method of controlling multiple mixing devices.

1.3 Objectives

The main objective of this project is to design a PLC-based sequential control system for three mixing devices using timers.

The specific objectives are:

1. To understand the principles of PLC-based industrial automation.
2. To design a ladder logic program for sequential control.
3. To assign suitable PLC inputs and outputs for the three mixing devices.
4. To start Mixer A after a delay of 7 seconds.
5. To start Mixer B 3.6 seconds after Mixer A starts.
6. To start Mixer C 5 seconds after Mixer B starts.
7. To ensure that all mixers remain ON until the stop condition occurs.
8. To implement a common stopping mechanism for all three mixers.
9. To reset the control sequence when the process is stopped.
10. To verify the sequential operation through PLC simulation or practical testing.

1.4 Scope of the Project

The scope of this project is limited to the design and implementation of a timer-based sequential control system for three mixing devices using a Programmable Logic Controller.

The system includes the following functions:

- Initiating the process through the Process Start input (X0).
- Activating Mixer A after a delay of 7 seconds.
- Activating Mixer B after an additional delay of 3.6 seconds.
- Activating Mixer C after an additional delay of 5 seconds.
- Maintaining the ON state of each activated mixer.
- Stopping all three mixers through the Process Stop input (X1) or master enable switch.
- Resetting the sequence to allow a new operating cycle.
- Verifying the sequence through simulation or hardware testing.

The project focuses on sequential starting and stopping operations. Advanced functions such as automatic ingredient measurement, temperature regulation, liquid-level monitoring, variable-speed control, and SCADA integration are outside the basic scope of the project. These features may be incorporated in future developments to improve the functionality and monitoring capabilities of the system.

1.5 Abbreviations and Symbols

Abbreviation / Symbol	Description
PLC	Programmable Logic Controller
CPU	Central Processing Unit
I/O	Input/Output
LD	Ladder Diagram
NO	Normally Open
NC	Normally Closed
X0	Process Start
X1	Process Stop
Y0	Mixer A Output
Y1	Mixer B Output
Y2	Mixer C Output
M0	Internal Relay / Sequence Enable
T0	Timer for Mixer A
T1	Timer for Mixer B
T2	Timer for Mixer C
s	Seconds
ms	Milliseconds

CHAPTER 2: LITERATURE SURVEY & PROPOSED WORK

2.1 Introduction

A literature survey helps identify existing techniques, technologies, and research related to the proposed project. In this project, the literature survey focuses on PLC-based sequential control, ladder logic programming, timer-based automation, and automatic mixing systems.

Programmable Logic Controllers (PLCs) are commonly used in industrial automation because they can execute programmed operations based on input conditions and control connected output devices. Ladder logic provides a graphical method of developing control sequences and is widely used for industrial machine control.

In sequential control systems, different operations are performed in a predefined order. Timers are used to introduce specific time delays between operations, allowing machines to start or stop at appropriate intervals. This approach reduces the need for manual timing and improves the consistency of industrial processes.

The proposed project applies these principles to a system consisting of three mixing devices. Mixer A starts after 7 seconds, Mixer B starts 3.6 seconds after Mixer A, and Mixer C starts 5 seconds after Mixer B. The system is designed so that all three mixers remain ON until the stop condition occurs.

The literature reviewed provides a technical foundation for developing the proposed control system using PLC timers, internal relays, and ladder logic.

2.2 Literature Review

The following research papers are relevant to PLC-based sequential control and automated mixing systems.

Table 1: Literature Review

Sr. No.	Research Paper / Authors	Year	Description	Relevance to Our Project
1	PLC Based Sequential Batch Process Control System — R. G. Jamkar, R. H. Chile and Y. S. Angal [1]	2006	The research presents a PLC-based system for controlling sequential batch processes using ladder logic and timers. *Taylor & Francis Online*	Supports the use of PLC timers to control the sequential starting of three mixing devices.
2	Fast Designing Ladder Diagram of Programmable Logic Controller for a Technological Process — Adam Szcześniak and Zbigniew Szcześniak [2]	2022	The paper explains methods for designing ladder diagrams for sequential technological processes. *ijet.pl*	Helps in developing the ladder logic required to control Mixers A, B and C.
3	Design of PLC Electrical Control System — Likai Ye [3]	2024	This study discusses PLC-based control of a liquid mixing system and the coordination of equipment during processing. *centurysciencepub.com*	Provides a related example of automated mixing using PLC technology.
4	Automation of a Pharmaceutical Batch Mixer Using PLC and SCADA — Augustine Kweku Mbeah and Erwin Normanyo [4]	2024	The study explores the automation and monitoring of a pharmaceutical batch mixer using PLC and SCADA technologies. *Science Publishing Group*	Provides ideas for future improvements, including process monitoring and advanced control.
5	PLC Based Liquid Mixing Device — Abhay Shukla et al. [5]	2024	The paper describes a PLC-based liquid mixing device for industrial applications, including process visualization using SCADA. *ijies.net*	Relates to the use of PLCs for automating mixing operations.

2.3 Methodology Used

The project follows a systematic methodology to design and implement the PLC-based sequential control system.

Step 1: Requirement Analysis

The operating requirements are identified, including the number of mixing devices, starting delays, input signals, output assignments, and stopping conditions.

Step 2: Hardware Selection

A suitable Programmable Logic Controller is selected as the main controller. Start and Stop inputs are used to initiate and terminate the process, while three PLC outputs control the respective mixer circuits.

Step 3: I/O Assignment

The PLC input/output assignments are defined as follows:

Address	Function
X0	Process Start
X1	Process Stop
Y0	Mixer A
Y1	Mixer B
Y2	Mixer C

Step 4: Ladder Logic Development

The ladder logic program is developed using PLC programming software. Internal relays and timer instructions are used to implement the required sequential operation.

Step 5: Timer Configuration

The timers are configured to achieve the specified delays:

- Timer T0: 7 seconds for Mixer A.
- Timer T1: 3.6 seconds for Mixer B.
- Timer T2: 5 seconds for Mixer C.

The timer settings must be selected according to the time base and timer instructions supported by the chosen PLC.

Step 6: Simulation and Testing

The ladder logic program is tested through PLC simulation or practical hardware testing. The test verifies that Mixer A starts after 7 seconds, Mixer B starts after another 3.6 seconds, and Mixer C starts after a further 5 seconds.

The stopping condition is also tested to confirm that all mixer outputs switch OFF when the process is stopped or the master enable is turned OFF.

Step 7: Result Analysis

The observed sequence is compared with the required operating sequence. Any timing or logic errors are corrected before the system is considered complete.

CHAPTER 3: SYSTEM DESIGN & IMPLEMENTATION

3.1 Components & Hardware

The PLC-based sequential control system consists of a Programmable Logic Controller, input devices, output devices, and timer instructions. These components work together to control the starting sequence of three mixing devices according to the specified time intervals.

The main components required for the proposed system are listed below.

Table 2: Components & Hardware

Sr. No.	Component / Hardware	Quantity
1	Mitsubishi FX5U PLC or equivalent PLC	1
2	PLC Programming Software	1
3	Process Start Push Button	1
4	Process Stop Push Button	1
5	Mixer A	1
6	Mixer B	1
7	Mixer C	1
8	PLC Timer Instructions	3
9	Motor Contactors and Overload Protection	As required
10	Connecting Wires and Power Supply	As required

3.2 Ladder Diagram

The ladder diagram represents the control logic used to operate the three mixing devices in a predefined sequence. The program uses input contacts, internal relays, timer instructions, and output coils to control the starting and stopping operations.

The input/output assignments are:

Address	Description
X0	Process Start
X1	Process Stop
Y0	Mixer A
Y1	Mixer B
Y2	Mixer C
M0	Internal sequence-enable relay
T0	Timer for Mixer A
T1	Timer for Mixer B
T2	Timer for Mixer C

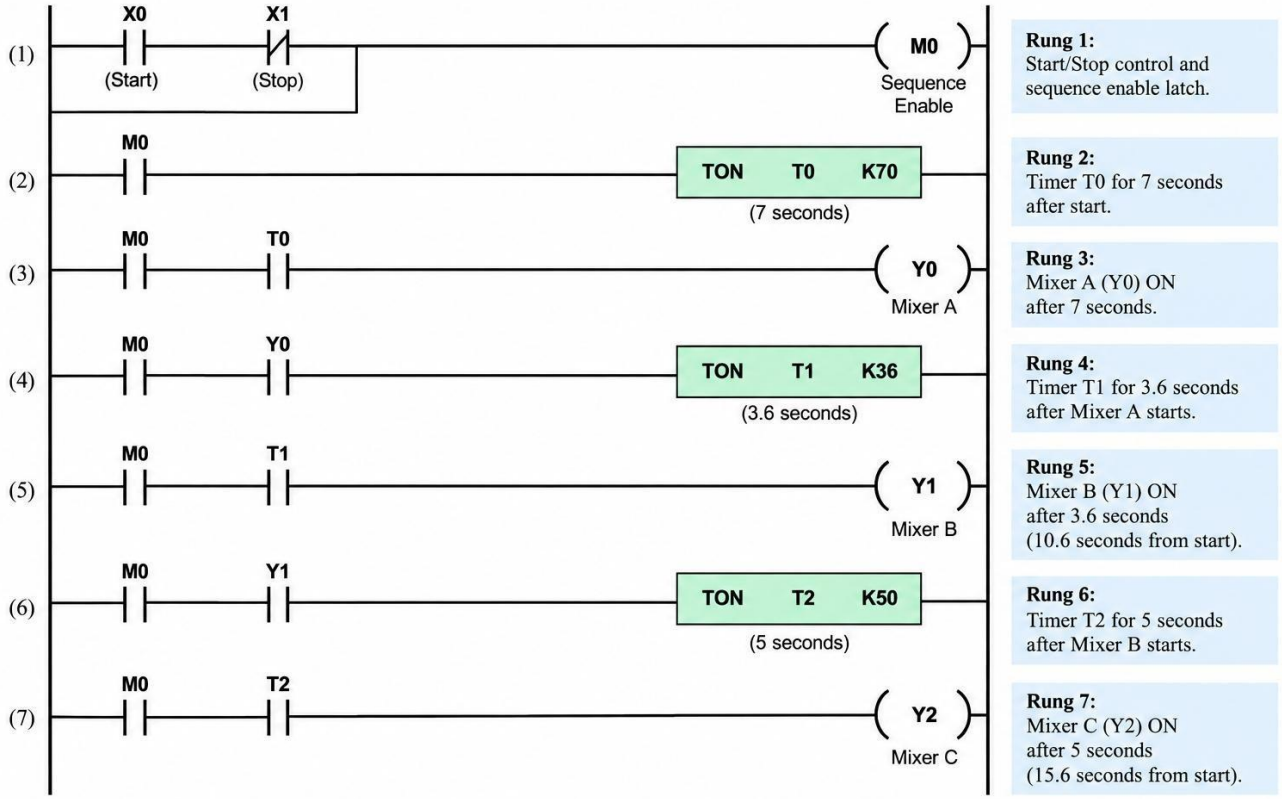


Figure 3.2.1: Ladder Diagram for Sequential Control of Three Mixing Devices

Figure 3.2.1: Sequential Ladder Logic Operation (conceptual representation).

Working of the Ladder Logic

1. When the Process Start input X0 is activated, the sequence-enable relay M0 is set, provided the stop and master-enable conditions permit operation.
2. Timer T0 begins counting. After 7 seconds, Mixer A connected to Y0 turns ON.
3. After Mixer A starts, timer T1 introduces an additional delay of 3.6 seconds. When this timer completes, Mixer B connected to Y1 turns ON.
4. After Mixer B starts, timer T2 introduces a further delay of 5 seconds. When this timer completes, Mixer C connected to Y2 turns ON.
5. Once activated, each mixer remains ON while the sequence is enabled.
6. When the Stop input is activated or the master enable is turned OFF, all mixer outputs turn OFF and the timers and sequence state reset.

3.3 Block Diagram

The block diagram illustrates the overall arrangement of the PLC-based sequential mixing system. The input devices provide the Start and Stop signals to the PLC. The PLC processes these signals and executes the timer-based control sequence to activate the three mixer outputs.

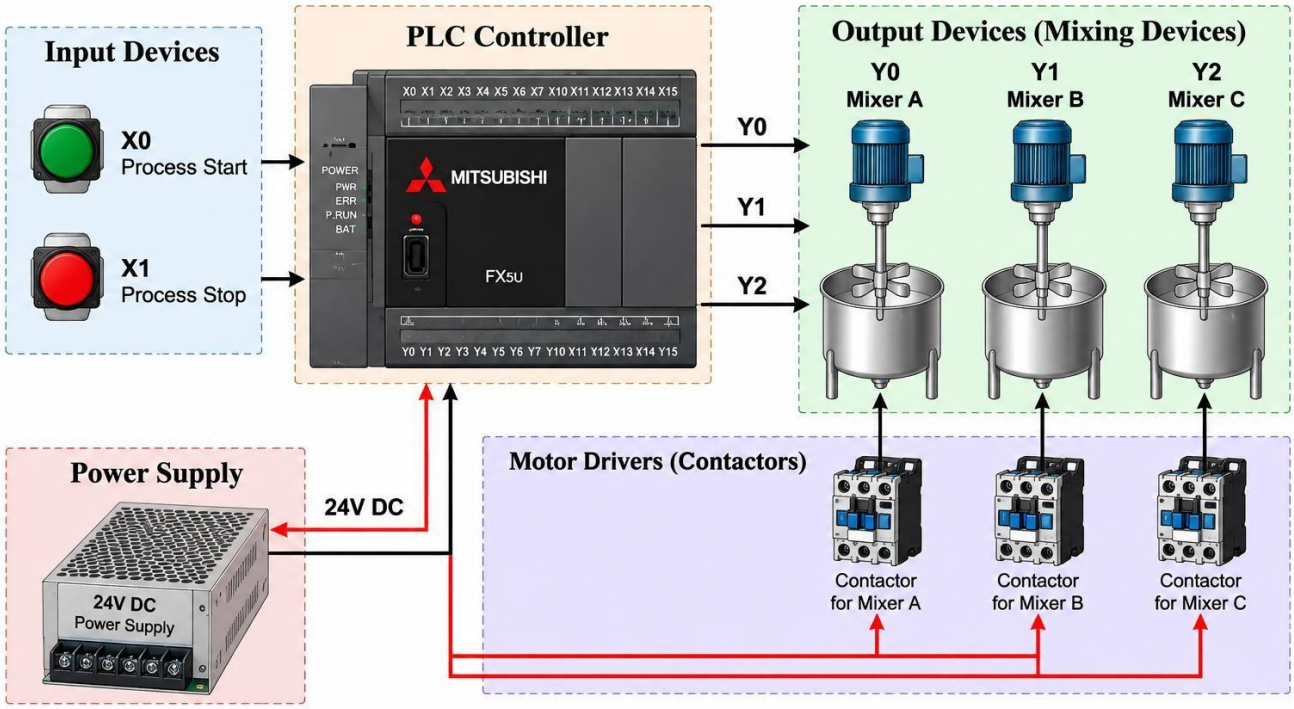


Figure 3.3.1: Block Diagram of the PLC-Based Sequential Mixing System

Working Principle

The system starts when the Process Start input X0 is activated. The PLC begins the first timer, and Mixer A starts after 7 seconds. Following the activation of Mixer A, Mixer B starts after a further 3.6 seconds. Mixer C starts after another 5 seconds following the activation of Mixer B.

Consequently, the three mixers start at 7 seconds, 10.6 seconds, and 15.6 seconds after the beginning of the process. All three mixers remain ON until the stop condition occurs. When the Stop input is activated or the master enable is turned OFF, the PLC deactivates all three outputs and resets the sequence.

CHAPTER 4: DESIGN & IMPLEMENTATION

4.1 Result

The proposed PLC-based sequential control system is designed to operate three mixing devices according to predefined time intervals. The system uses timer instructions and ladder logic to control the starting sequence of Mixer A, Mixer B, and Mixer C.

When the Process Start input (X0) is activated, the control sequence begins. Mixer A, connected to output Y0, turns ON after 7 seconds. Mixer B, connected to output Y1, turns ON after an additional delay of 3.6 seconds, resulting in an elapsed time of 10.6 seconds from the beginning of the process. Mixer C, connected to output Y2, turns ON after a further delay of 5 seconds, resulting in an elapsed time of 15.6 seconds.

Once activated, all three mixers remain ON until the Process Stop input (X1) is activated or the master enable switch is turned OFF. Under this condition, all mixer outputs are switched OFF and the sequence is reset for the next operation.

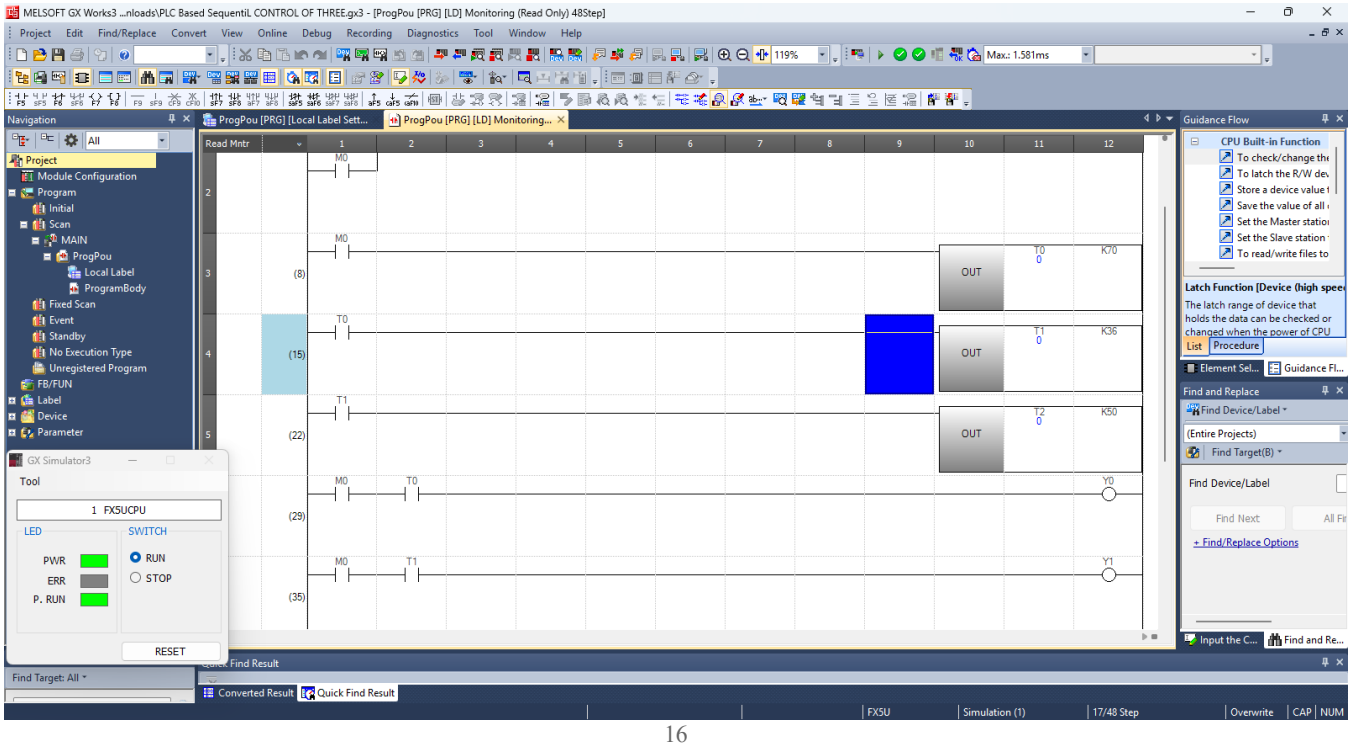
The expected operation of the system is summarized below.

Table 3: Expected Sequence of Operation

Sr. No.	Time from Start	Operation	Output Status
1	0 seconds	Process starts	All mixers OFF
2	7 seconds	Mixer A starts	Y0 ON
3	10.6 seconds	Mixer B starts	Y0, Y1 ON
4	15.6 seconds	Mixer C starts	Y0, Y1, Y2 ON
5	When Stop is activated	Process terminates	All mixers OFF

The table represents the expected sequence based on the specified timer settings. The actual operation should be verified using PLC simulation or practical hardware testing.

4.2 Photos



4.3 Flow Diagram

The flow diagram represents the sequence of operations performed by the PLC during the mixing process.

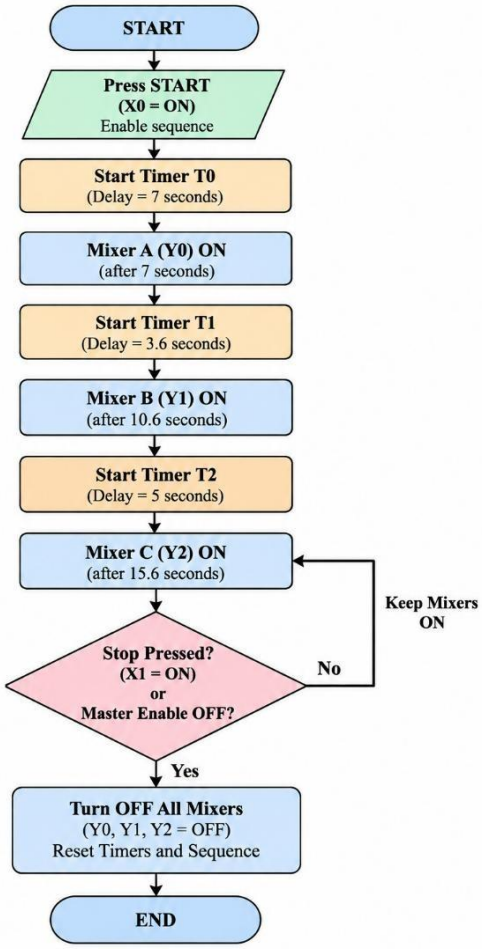


Figure 4.3.1: Flow Diagram of Sequential Mixing Operation

Figure 4.3.1: Flow Diagram of Sequential Mixing Operation

The process begins when X0 is activated. The PLC executes the timer sequence and switches ON the mixers at their specified intervals. After all three mixers are running, they remain ON while the system is enabled. When the stop condition occurs, the PLC switches OFF all mixer outputs and resets the sequence.

CHAPTER 5: APPLICATION & FUTURE SCOPE

5.1 Applications

The PLC-based sequential control system for three mixing devices can be used in various industrial applications where machines must start at predefined time intervals. The use of PLC timers and ladder logic enables automated operation, reduces manual intervention, and provides consistent control of the equipment.

1. Chemical Processing Industries

The system can be used in chemical processing plants where mixing equipment must be activated in a predefined sequence. Timer-based control helps coordinate the starting of multiple mixers during processing operations.

2. Food Processing Industries

In food processing industries, mixing devices are used for blending ingredients and preparing different food products. The proposed system can automate the sequential starting of mixing equipment and improve operational consistency.

3. Pharmaceutical Industries

Pharmaceutical manufacturing involves mixing different materials during the preparation of products. A PLC-based sequential control system can provide timed coordination of mixing equipment. Actual pharmaceutical applications require process validation and appropriate safety controls.

4. Manufacturing and Production Lines

The system can be applied to manufacturing processes where multiple machines need to start at specific time intervals. Automatic sequencing reduces the need for operators to manually coordinate each machine.

5. Batch Processing Systems

Batch processing systems often require equipment to operate in a predefined order. The proposed PLC system can be adapted to control mixing devices during batch preparation by configuring suitable timer intervals.

6. Industrial Automation Training

The project is useful for understanding PLC programming, ladder logic, timer instructions, input/output assignments, and sequential machine control. It can be used as a practical demonstration in industrial automation laboratories.

5.2 Future Scope

The proposed system provides a basic foundation for sequential control of three mixing devices. Its functionality can be improved by incorporating additional automation technologies and monitoring features.

1. Human-Machine Interface (HMI) Integration

An HMI can be added to display the operating status of each mixer, timer values, and system conditions. It can also allow authorized operators to monitor and control the process through a graphical interface.

2. Sensor Integration

Sensors such as temperature sensors, level sensors, and material-detection sensors can be integrated into the system. These sensors can provide additional input conditions and help the PLC respond to changes in the processing environment.

3. Adjustable Timer Settings

The timer values can be made adjustable so that operators can configure the starting delays according to different process requirements. Suitable access controls and validated operating limits should be applied.

4. Fault Detection and Alarm System

Motor overload feedback and fault sensors can be incorporated to detect abnormal operating conditions. The PLC can stop the sequence and activate an alarm when a fault is detected.

5. SCADA Integration

Supervisory Control and Data Acquisition (SCADA) technology can be used to monitor the mixing process remotely. It can display operating status, record events, and help operators analyze process performance.

6. Expansion to Multiple Mixing Devices

The system can be expanded to control more than three mixing devices by adding further PLC outputs, timer stages, and suitable interlocking logic. This would make the system adaptable to larger industrial processing lines.

7. Data Logging and Process Monitoring

A data-logging facility can be introduced to record mixer activation times, operating duration, and fault events. The collected information can help improve process monitoring and maintenance planning.

Summary

The proposed PLC-based sequential control system has applications in industries that require timed coordination of multiple mixing devices. Future improvements such as HMI integration, sensor-based control, fault detection, and SCADA monitoring can enhance its flexibility, reliability, and suitability for more advanced industrial automation applications.

CHAPTER 6: CONCLUSION & REFERENCES

6.1 Conclusion

The project, "PLC-Based Sequential Control of Three Mixing Devices Using Timers," focuses on designing a timer-based control system for the sequential operation of three mixing devices using a Programmable Logic Controller (PLC). The system is designed to automate the starting sequence of the mixers according to predefined time intervals.

The operation begins when the Process Start input (X0) is activated. Mixer A, connected to output Y0, starts after a delay of 7 seconds. Mixer B, connected to output Y1, starts after an additional delay of 3.6 seconds, resulting in a total elapsed time of 10.6 seconds from the beginning of the process. Mixer C, connected to output Y2, starts after a further delay of 5 seconds, resulting in a total elapsed time of 15.6 seconds.

This project demonstrates the application of PLC-based sequential control in industrial automation. It provides an understanding of timer configuration, ladder logic programming, input/output assignment, and the coordination of multiple devices. The system can be adapted for applications in manufacturing, food processing, chemical industries, and other processes that require timed machine operation.

The proposed design provides a foundation for further improvements, such as HMI integration, sensor-based control, fault detection, and SCADA monitoring. The actual performance of the system should be confirmed through PLC simulation or hardware testing before it is considered a verified implementation.

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